

HW # 12 - David Alvarado

1) Laminar flow: $V_z = V_z^{\max} \left(1 - \left(\frac{r}{R} \right)^2 \right)$ $\langle V \rangle = \frac{Q}{A} = \frac{\int_0^{2\pi} \int_0^R V_z r dr d\theta}{\int_0^{2\pi} \int_0^R r dr d\theta} = \frac{1}{2} V_z^{\max}$

Turbulent flow: $V_z = V_z^{\max} \left(1 - \frac{r}{R} \right)^{1/4}$

Laminar flow $\left[\frac{\langle V_z^2 \rangle}{\langle V_z \rangle^2} \right]^{(3)}$

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① $\langle V_z^2 \rangle = ?$
 $\left[V_z = V_z^{\max} \left(1 - \left(\frac{r}{R} \right)^2 \right) \right]^2$

$V_z^2 = V_z^{\max 2} \left(1 - \left(\frac{r}{R} \right)^2 \right)^2$

$\langle V_z^2 \rangle = V_z^{\max 2} \left(1 - \left(\frac{r}{R} \right)^2 - \left(\frac{r}{R} \right)^2 + \left(\frac{r}{R} \right)^4 \right)$
 $= V_z^{\max 2} \left(1 - 2 \left(\frac{r}{R} \right)^2 + \left(\frac{r}{R} \right)^4 \right)$

$\langle V_z^2 \rangle = \frac{\int_0^{2\pi} \int_0^R V_z^2 \cdot r dr d\theta}{\int_0^{2\pi} \int_0^R r dr d\theta}$

$= \frac{\int_0^{2\pi} \int_0^R \left(V_z^{\max 2} \left(1 - 2 \left(\frac{r}{R} \right)^2 + \left(\frac{r}{R} \right)^4 \right) \right) r dr d\theta}{2\pi \cdot \frac{R^2}{2} \cdot 1}$

$= \frac{\int_0^{2\pi} d\theta \cdot V_z^{\max 2} \int_0^R \left(r - 2 \frac{r^3}{R^2} + \left(\frac{r^5}{R^4} \right) \right) dr}{\pi R^2}$

$= \frac{2\pi (V_z^{\max})^2 \left[\frac{r^2}{2} - \frac{2}{R^2} \left(\frac{r^4}{4} \right) + \frac{1}{R^4} \left(\frac{r^6}{6} \right) \right]_0^R}{\pi R^2}$

$= \frac{2 (V_z^{\max})^2}{R^2} \left[\frac{R^2}{2} - \frac{2}{R^2} \left(\frac{R^4}{4} \right) + \frac{1}{R^4} \left(\frac{R^6}{6} \right) \right]$

$= \frac{2 (V_z^{\max})^2}{R^2} \left(\frac{R^2}{2} - \frac{R^2}{2} + \frac{R^2}{6} \right)$

$\langle V_z^2 \rangle = \frac{1}{3} (V_z^{\max})^2$

$\Rightarrow \textcircled{3} \frac{\langle V_z^2 \rangle}{\langle V_z \rangle^2} = \frac{\left(\frac{1}{3} (V_z^{\max})^2 \right)}{\left(\frac{1}{4} (V_z^{\max})^2 \right)} = \frac{1}{3} \cdot \frac{4}{1} = \frac{4}{3}$

$\frac{\langle V_z^2 \rangle}{\langle V_z \rangle^2} = \frac{4}{3}$

② $\langle V_z^2 \rangle = ?$
 $V_z = V_z^{\max} \left(1 - \left(\frac{r}{R} \right)^2 \right) = V_z^{\max} \left(1 - \frac{r^2}{R^2} \right)$

$\langle V_z \rangle = \frac{\int_0^{2\pi} \int_0^R V_z^{\max} \left(1 - \frac{r^2}{R^2} \right) r dr d\theta}{\int_0^{2\pi} \int_0^R r dr d\theta}$

$= \frac{\int_0^{2\pi} d\theta \cdot V_z^{\max} \int_0^R \left(r - \frac{r^3}{R^2} \right) dr}{\pi R^2}$

$= \frac{2 V_z^{\max}}{R^2} \left[\frac{r^2}{2} - \frac{1}{R^2} \left(\frac{r^4}{4} \right) \right]_0^R$

$= \frac{2 V_z^{\max}}{R^2} \left[\frac{R^2}{2} - \frac{1}{R^2} \left(\frac{R^4}{4} \right) \right]$

$\frac{2 V_z^{\max}}{R^2} \left(\frac{R^2}{2} \right)$

$V_z^{\max} / 2 = \langle V_z \rangle$

$\langle V_z^2 \rangle = \left(\frac{V_z^{\max}}{2} \right)^2$

$\langle V_z^2 \rangle = \frac{1}{4} (V_z^{\max})^2$

Turbulent flow

$$\frac{\langle V_z^2 \rangle}{\langle V_z \rangle^2} = ? \quad] ③$$

$$① \langle V_z \rangle = ?$$

$$V_z = V_{\max} \left(1 - \frac{r}{R}\right)^{1/4}$$

$$V_z^2 = V_{\max}^2 \left(1 - \frac{r}{R}\right)^{1/2}$$

$$\langle V_z^2 \rangle = \int_0^{2\pi} \int_0^R V_{\max}^2 \left(1 - \frac{r}{R}\right)^{1/2} r dr d\theta$$

$$\int_0^{2\pi} \int_0^R r dr d\theta$$

$$= \frac{2\pi V_{\max}^2}{\pi R^2} \int_0^R \left(1 - \frac{r}{R}\right)^{1/2} \cdot r dr$$

Integral
calc. $\left(\int\right) = \frac{2V_{\max}^2}{\pi R^2} \left(\frac{49}{144} R^2\right)$

$$\langle V_z^2 \rangle = \frac{49}{72} V_{\max}^2$$

$$② \langle V_z \rangle^2 = ?$$

$$V_z = V_{\max} \left(1 - \frac{r}{R}\right)^{1/4}$$

$$\langle V_z \rangle = \frac{\int_0^{2\pi} \int_0^R V_{\max} \left(1 - \frac{r}{R}\right)^{1/4} r dr d\theta}{\int_0^{2\pi} \int_0^R r dr d\theta}$$

$$\langle V_z \rangle = \frac{2\pi V_{\max} \int_0^R \left(1 - \frac{r}{R}\right)^{1/4} r dr}{\pi R^2}$$

$$\langle V_z \rangle = \frac{2V_{\max}}{\pi R^2} \left(\frac{49}{120} R^2\right)$$

$$\langle V_z \rangle = \frac{49}{60} V_{\max}$$

$$\langle V_z \rangle^2 = \frac{2401}{3600} V_{\max}^2$$

$$\Rightarrow \frac{\langle V_z^2 \rangle}{\langle V_z \rangle^2} = \frac{\frac{49}{72} V_{\max}^2}{\frac{2401}{3600} V_{\max}^2} = \frac{49}{72} \cdot \frac{3600}{2401} = \frac{50}{49}$$

$$\frac{\langle V_z^2 \rangle}{\langle V_z \rangle^2} = \frac{50}{49}$$

$$4) \frac{V_z}{\bar{V}_{z, \max}} = \left(1 - \frac{r}{R}\right)^{1/n}$$

$$\bar{V}_z = \bar{V}_{z, \max} \left(1 - \frac{r}{R}\right)^{1/n}$$

$$\frac{\langle \bar{V}_z \rangle}{\bar{V}_{z, \max}} = ?$$

$$\langle \bar{V} \rangle = \frac{Q}{A} = \frac{\int_0^R \int_0^{2\pi} \bar{V}_z r dr d\theta}{\int_0^R \int_0^{2\pi} r dr d\theta}$$

$$① \langle \bar{V}_z \rangle = ?$$

$$\bar{V}_z = \bar{V}_{z, \max} \left(1 - \frac{r}{R}\right)^{1/n}$$

$$\langle \bar{V} \rangle = \frac{\int_0^R \int_0^{2\pi} \bar{V}_{z, \max} \left(1 - \frac{r}{R}\right)^{1/n} r dr d\theta}{\int_0^R \int_0^{2\pi} r dr d\theta}$$

$$\langle \bar{V}_z \rangle = \frac{2\pi \bar{V}_{z, \max}}{\pi R^2} \int_0^R \left(1 - \frac{r}{R}\right)^{1/n} r dr$$

Integral calculator

$$\langle \bar{V}_z \rangle = \frac{2\bar{V}_{z, \max}}{R^2} \left(R \left(-R \frac{n}{2n+1} + R \frac{n}{n+1} \right) \right)$$

$$= \frac{2\bar{V}_{z, \max}}{R^2} \left(R^2 \left(\frac{-n(n+1) + (2n+1)n}{(2n+1)(n+1)} \right) \right)$$

$$= 2\bar{V}_{z, \max} \left(\frac{n^2}{(2n+1)(n+1)} \right)$$

$$= 2\bar{V}_{z, \max} \left(\frac{n^2}{(2n+1)(n+1)} \right)$$

$$\langle \bar{V} \rangle = \frac{2n^2}{(2n+1)(n+1)} \cdot \bar{V}_{z, \max}$$

$$② \frac{\langle \bar{V}_z \rangle}{\bar{V}_{z, \max}} = \frac{2n^2}{(2n+1)(n+1)}$$

$$\frac{\langle \bar{V}_z \rangle}{\bar{V}_{z, \max}} = \frac{2n^2}{(2n+1)(n+1)}$$